

# Glance ML Internship Assignment: Multimodal Fashion & Context Retrieval

## 1. Project Overview

The goal of this assignment is to build an intelligent search engine that can retrieve specific images from a diverse database based on natural language descriptions. The system needs to understand not just "what" someone is wearing, but "where" they are and the "vibe" of their attire.

## 2. The Dataset

You are responsible for sourcing or simulating a dataset (minimum 500–1,000 images). The data must contain variations across three primary axes:

- **Environment:** Office interiors, urban streets, parks, and home settings.
- **Clothing Types:** Formal (blazers, button-downs), Casual (hoodies, t-shirts), and Outerwear, etc
- **Color Theory:** A wide palette of garment colors.

One such dataset is fashionpedia's [dataset](#) or you can download it [here](#)

## 3. Technical Requirements

Your submission must consist of two distinct workflows, ideally hosted in 1 GitHub repository in different directories or files, clearly defined modules:

### Part A: The Indexer

- **Feature Extraction:** Process the raw images into a searchable format.
- **Vector Storage:** Implement a solution to store these representations efficiently (avoiding simple filename keyword matching).

### Part B: The Retriever (The "Query")

- **Search Logic:** Create a script that accepts a natural language string and returns the top \$k\$ matching images.
- **Context Awareness:** The system should handle multi-attribute queries (e.g., color + clothing type + location).

Focus more on the ML logic aspect of it, rather than the engg work of indexing, for example - if you chose to pick a Vector DB, pick the easiest and most convenient one instead of

spending time and code on rewriting your own version. You are assessed first and foremost on ML logic.

**Hint:** While architectures like CLIP provide a strong baseline for zero-shot retrieval, they often struggle with compositionality (e.g., distinguishing "red shirt with blue pants" from "blue shirt with red pants") and fine-grained fashion attributes. The expected solution should be better than vanilla application of CLIP with focus on making it work for fashion based retrieval.

## 4. Evaluation Queries

Your system will be judged on its ability to accurately return images for the following prompts:

1. **Attribute Specific:** *"A person in a bright yellow raincoat."*
2. **Contextual/Place:** *"Professional business attire inside a modern office."*
3. **Complex Semantic:** *"Someone wearing a blue shirt sitting on a park bench."*
4. **Style Inference:** *"Casual weekend outfit for a city walk."*
5. **Compositional:** *"A red tie and a white shirt in a formal setting."*

## 5. Submission Deliverables

A single PDF that contains

1. **Approaches:** Possible ways to solve this problem, tradeoffs, what's good and when
2. **Short Write-up on Chosen Approach:** A brief explanation of your chosen architecture & how it handles fashion queries
3. **Codebase (GitHub) Link:** Clean, documented code for both the indexing and retrieval pipelines.
4. **Approaches for future work**
  - a. How to extend this solution for adding locations (cities, places) and weather
  - b. How to improve precision

## 6. What We Are Looking For

1. **Thoughtful solution:**
  - a. Getting a solution in the age of AI is easy - coding it successfully, understanding the chosen approach (model, arch), what are its shortcomings & how to address that are sought
  - b. How well does it work for fashion based queries?
2. **Modular Code:** Is your logic separated from your data?
3. **Scalability:** Would your retrieval logic work if the dataset grew to 1 million images?
4. **Zero-Shot Capability:** How well does the system handle descriptions it hasn't seen explicitly in a training label?